

IS6400 Business Data Analytics

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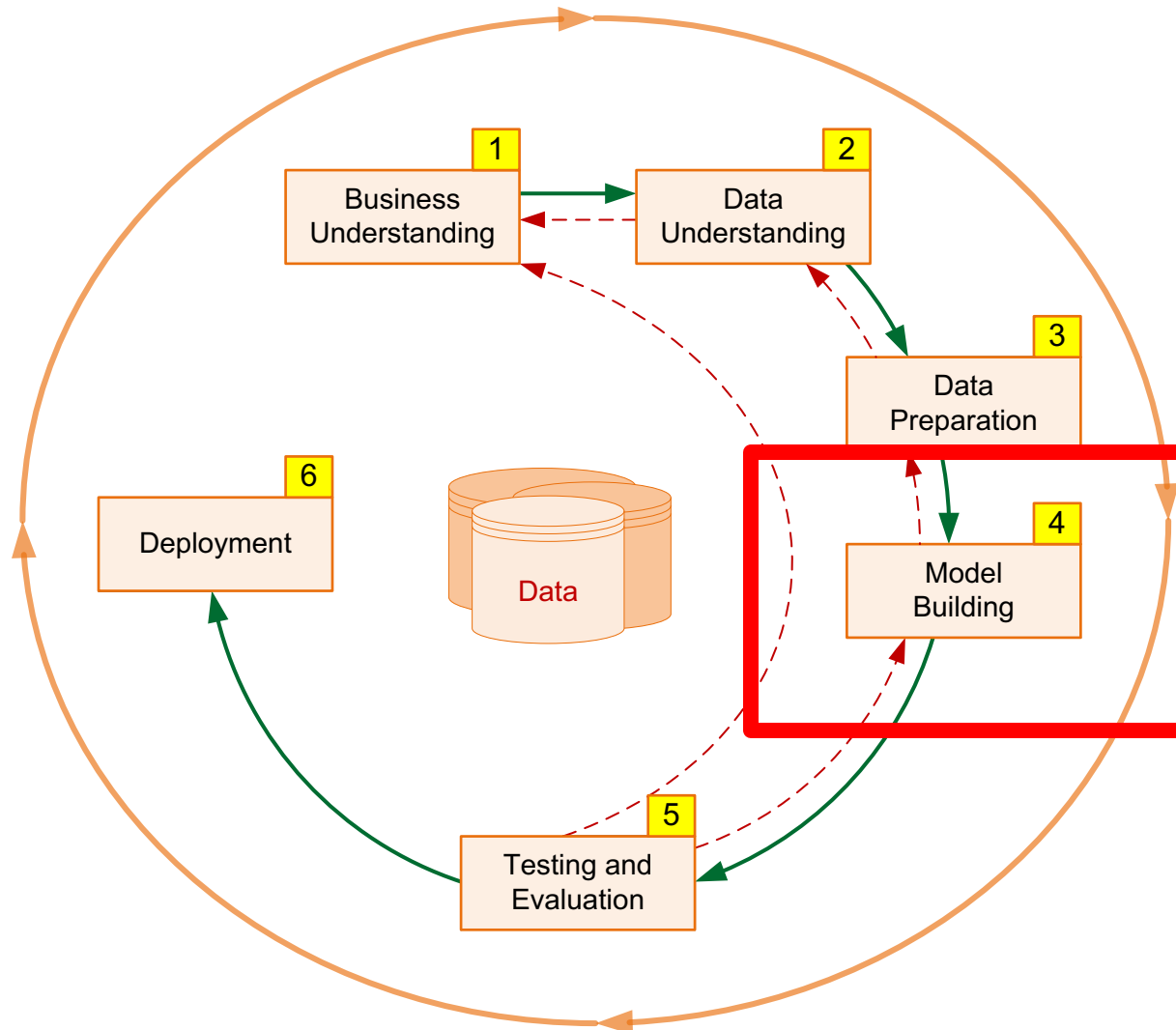
Lecture 2

- ☒ **Predictive Analytics**
 - ☒ **Linear Regression**
 - ☐ Parameter Training
 - ☐ Overfitting

Reading Material:

- ☐ **HM Chapter 4**

Predictive Analytics



Predictive Modeling

- ❖ Managerial decisions are often based on the relationship between two or more variables
- ❖ Sometimes a manager will rely on intuition to judge how two variables are related
- ❖ If data can be obtained, a statistical procedure called **regression analysis** can be used to develop an equation showing how the variables are related
- ❖ Linear regressions can be used in business to evaluate trends and make estimates or forecasts

Some Business Applications Using Linear Regression

❖ Evaluating trends and sales estimates

- A marketing manager attempts to predict sales for a given level of advertising expenditures through various media channels (e.g. TV, newspapers, social media)

❖ Analyzing the impact of price changes

- A company changes the price on a certain product several times and wants to predict the quantity it sells for each of the price levels

❖ Assessing risks

- A health insurance company wants to predict the number of claims per customer against age

Terminologies for Prediction

- ❖ Identify what to be predicted and potential predictors

Statisticians say	Data Miners say	Social Scientists say	They all Mean
Independent Variables	Predictor Variables	Explanatory Variables	Stuff you can use to predict the target
Dependent Variable	Target Variable	Response Variables	The thing you want to predict

Dependent vs. Independent Variables

❖ Evaluating trends and sales estimates

- A marketing manager attempts to predict sales (**dependent variable**) for a given level of advertising expenditures through various media channels (**independent variable**)

❖ Analyzing the impact of price changes

- A company changes the price on a certain product several times and wants to predict the quantity it sells (**dependent variable**) for each of the price levels (**independent variable**)

❖ Assessing risks

- A health insurance company wants to predict the number of claims (**dependent variable**) per customer against age (**independent variable**)

example - 1

Business Problem:

Will people be happier if GDP reaches 60000 per capital?

1. Transfer the business problem to a BDA task

This is a regression task

2. Study the data related to the problem

Find training data (data with GDP VS Happiness.)

Table 1-1. Does money make people happier?

Country	GDP per capita (USD)	Life satisfaction
Hungary	12,240	4.9
Korea	27,195	5.8
France	37,675	6.5
Australia	50,962	7.3
United States	55,805	7.2

example - 1

Business Problem:

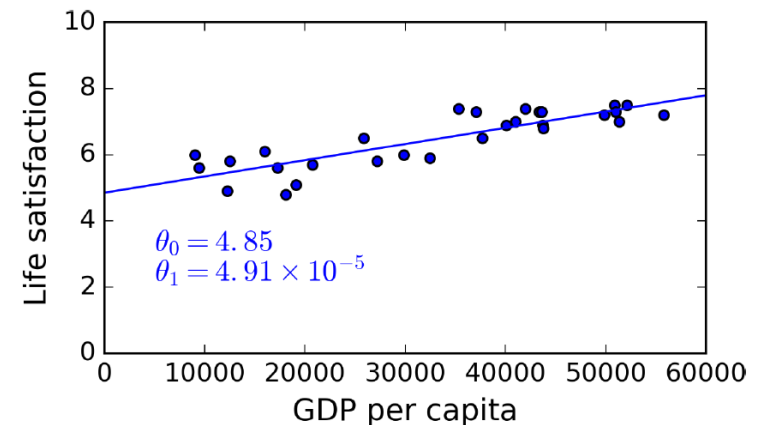
Will people be more happier if GDP reaches 60000 per capital?

3. Select a model and train it using the data

Table 1-1. Does money make people happier?

Country	GDP per capita (USD)	Life satisfaction
Hungary	12,240	4.9
Korea	27,195	5.8
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Train



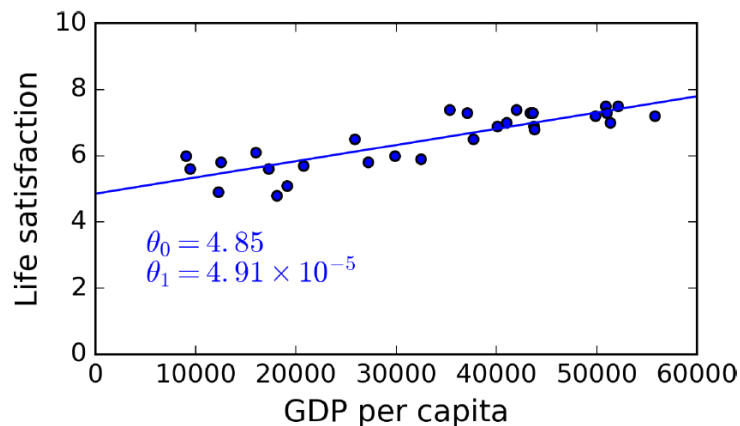
example - 1

Business Problem:

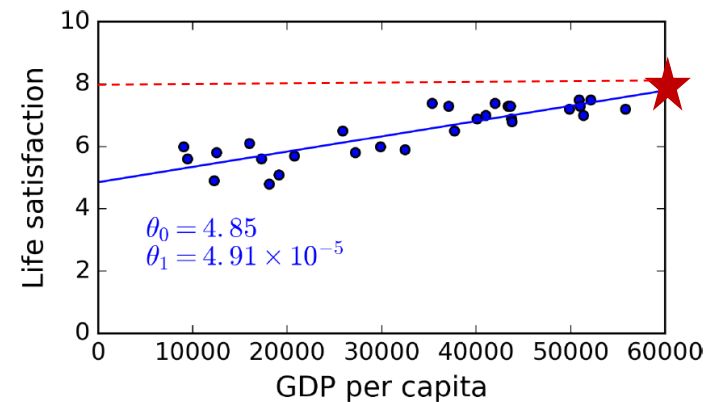
Will people be happier if GDP reaches 60000 per capital?

4. Implement the model for future data

Increase GDP to 60000?



Implement



An Example of Linear Regression

- ❖ Regression: Prediction based on historical data
- ❖ **Problem:** I know current egg prices in California. Can I predict egg sales for the future weeks in California?
- ❖ **I have:** Two-year historical database
 - Independent variable: egg prices
 - Dependent variable: weekly sales cases of eggs
- ❖ Step one: look at the raw historical data...

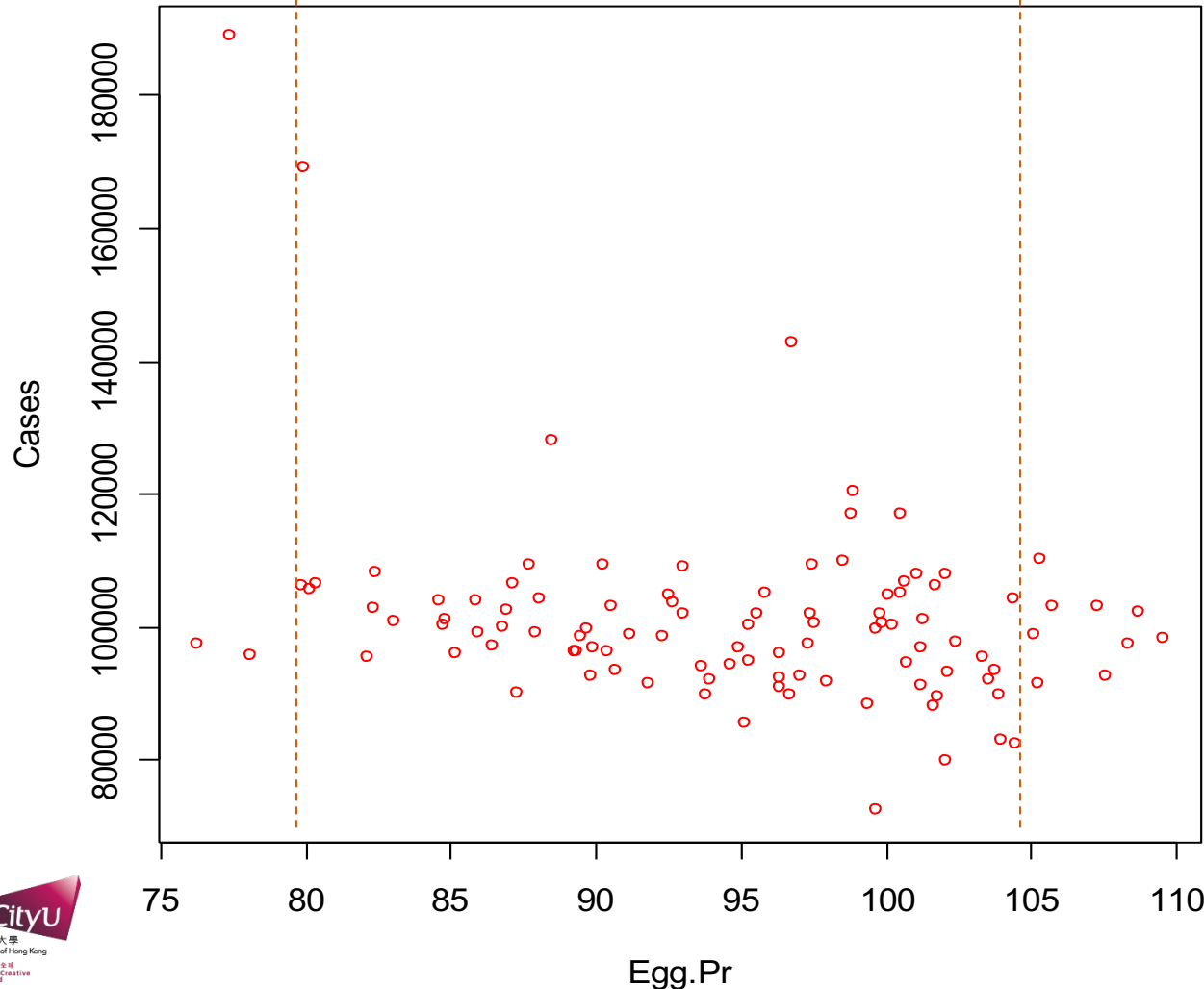
Historical Data

Cases Sold / Week	Egg Prices
96343	90.42
96345	89.33
96928	89.89
93519	90.71
99032	85.99
91539	91.83
89969	87.29
90859	96.36
99697	99.71
88350	99.38

Cases Sold / Week	Egg Prices
100383	97.53
94415	100.77
91813	98
100466	99.89
96783	101.26
91008	101.26
100324	100.28
106628	100.69
98892	105.16
98252	109.55

Scatterplot Representation

- ❖ A linear model that could be used for prediction



Prediction: At
\$.80, how many
cases would
you predict
would be sold?
At \$1.05?

Scatterplot to Algebra

❖ We want:

- A formula that we can plug the egg prices into and it will give us the predicted case sales
- And it will give us the error range in the predicted case sales
- First try: Use the simplest formula around: a linear relation:

$$\text{Case Sales} = a + b * \text{Egg Prices}$$

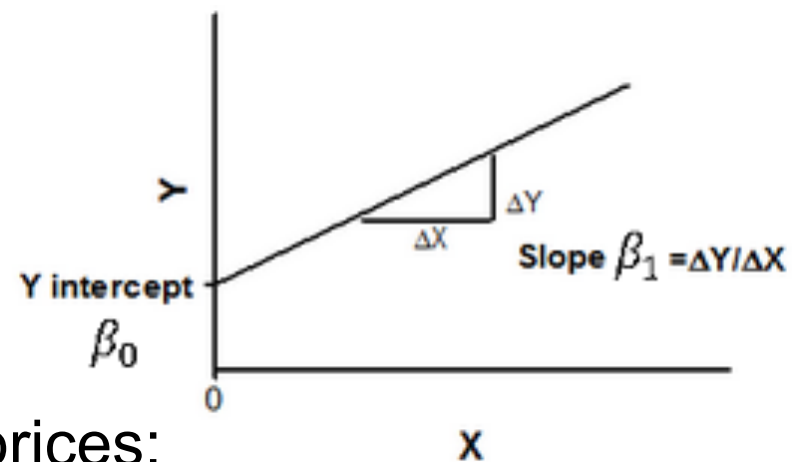
- Use the historical data to calibrate (estimate the values of a and b) this linear model

Simple Linear Regression

- ❖ To predict the value of one dependent variable on the basis of another independent variable

- Dependent Variable: **Y**
- Independent Variable: **X**

$$Y = \beta_0 + \beta_1 X$$



- ❖ Regression case sales on egg prices:

$$\text{Case sales} = \beta_0 + \beta_1 \times \text{Egg Price} + \varepsilon$$

- β_0 : Constant or intercept
- β_1 : the change of sales when the egg price increases 1 unit (i.e. Slope – a measure of how much egg case sales changes due to a given change in egg price, or how sensitive the value of Y is to changes in X)
- ε : random error

Simple Linear Regression

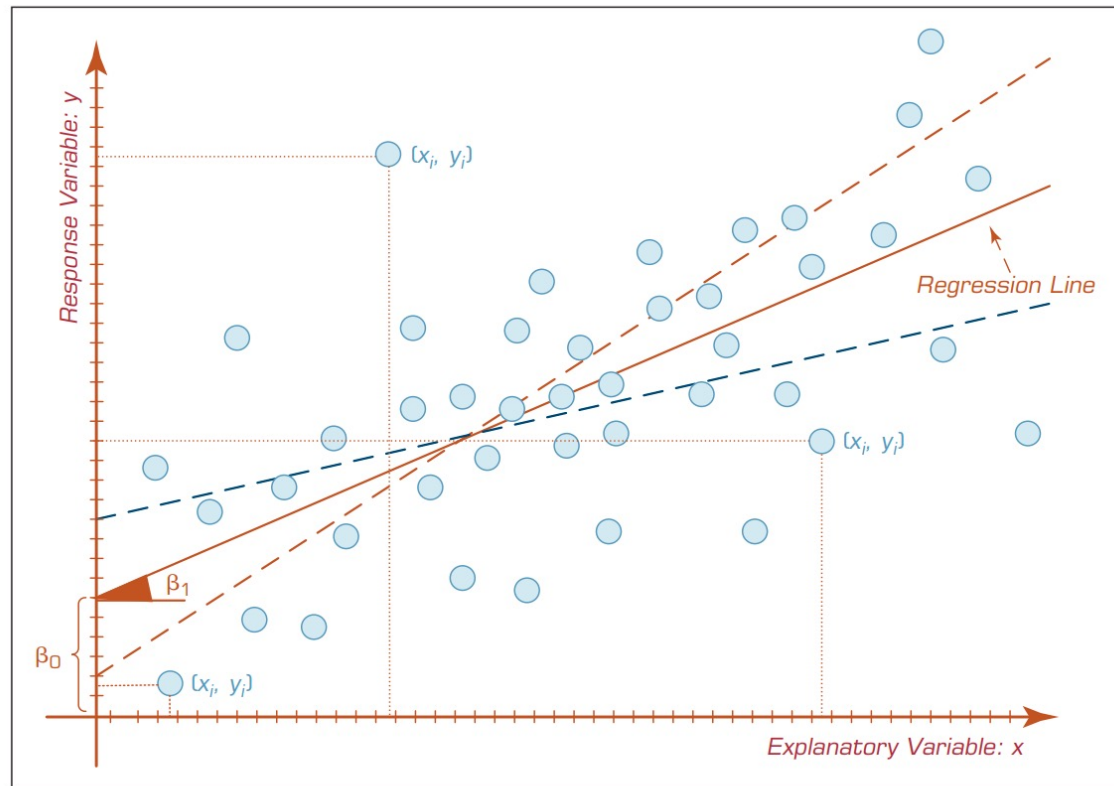


FIGURE 2.13 A Scatter Plot and a Linear Regression Line.

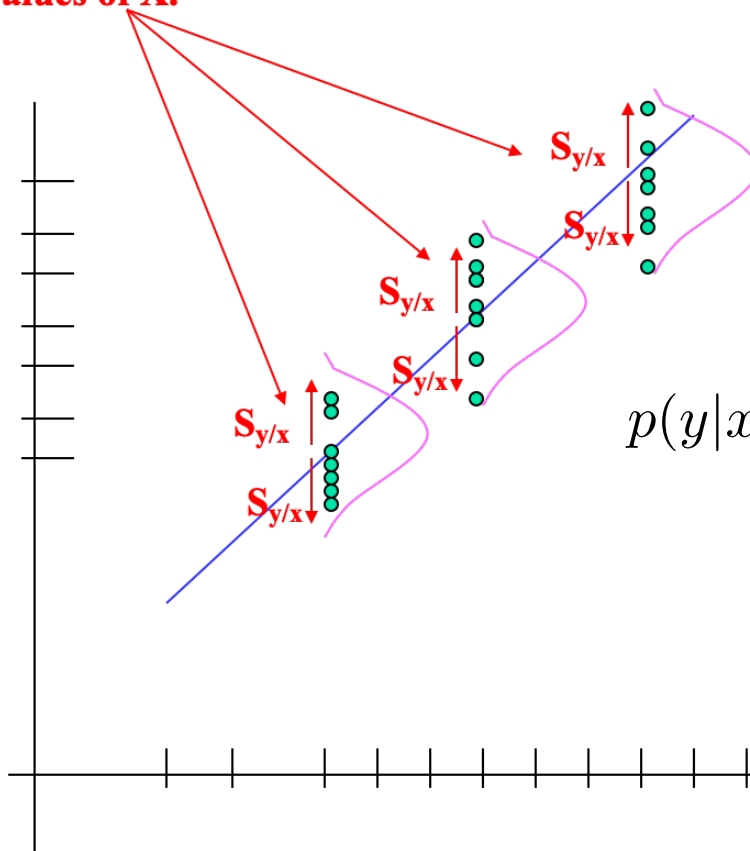
Prediction

If you know something about X , this knowledge helps you predict something about Y . (Sound familiar?...sound like conditional probabilities?)

Assumptions (or the fine print)

- ❖ Linear regression assumes that...
 - ❧ 1. The relationship between X and Y is linear
 - ❧ 2. Y is distributed normally at each value of X
 - ❧ 3. The variance of Y at every value of X is the same (homogeneity of variances)
 - ❧ 4. The observations are independent

The standard error of Y given X is the average variability around the regression line at any given value of X. It is assumed to be equal at all values of X.

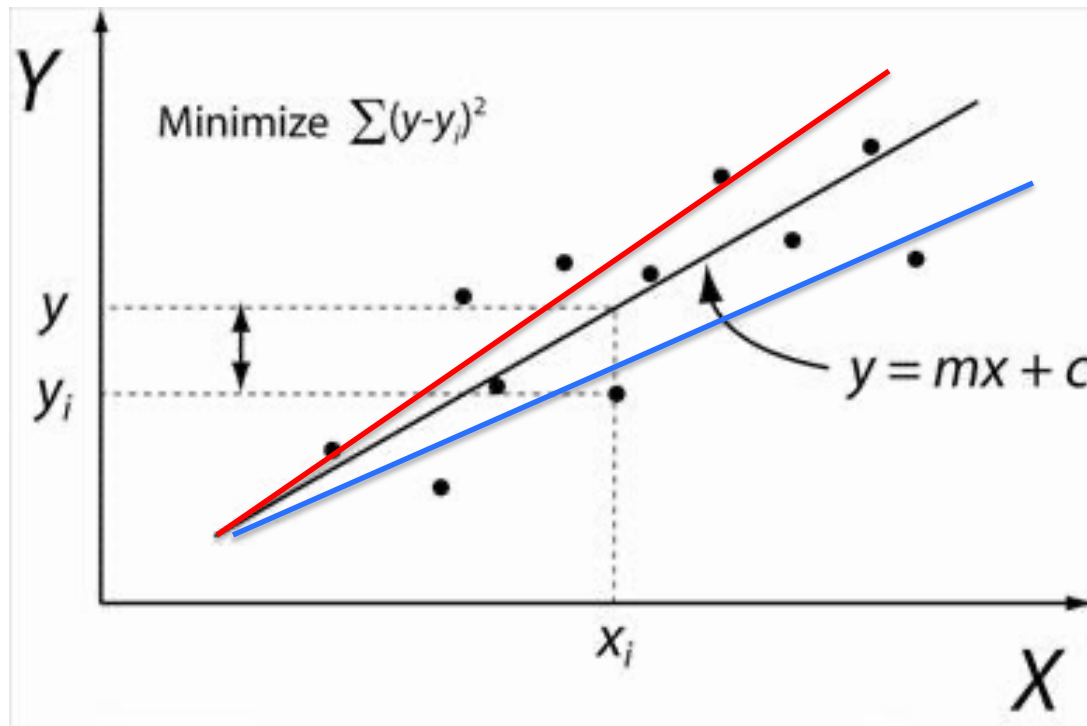


$$Y \sim N(\theta^T x, \sigma^2)$$

$$p(y|x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - \theta^T x)^2}{2\sigma^2}\right)$$

Regression Curve Fitting

- ❖ The values of $\beta_0, \beta_1 \dots$ are selected such that they provide the best fit, or the minimal errors between the data and the model (i.e. the sum of difference between actual and predicted values)



The Output

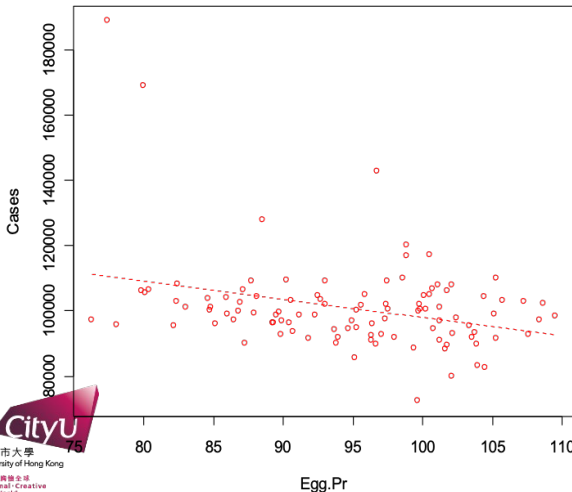
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
β_0	153,414.5	15,992.9	9.593	5.96e-16 ***
β_1	-553.9	168.2	-3.293	0.00136 **

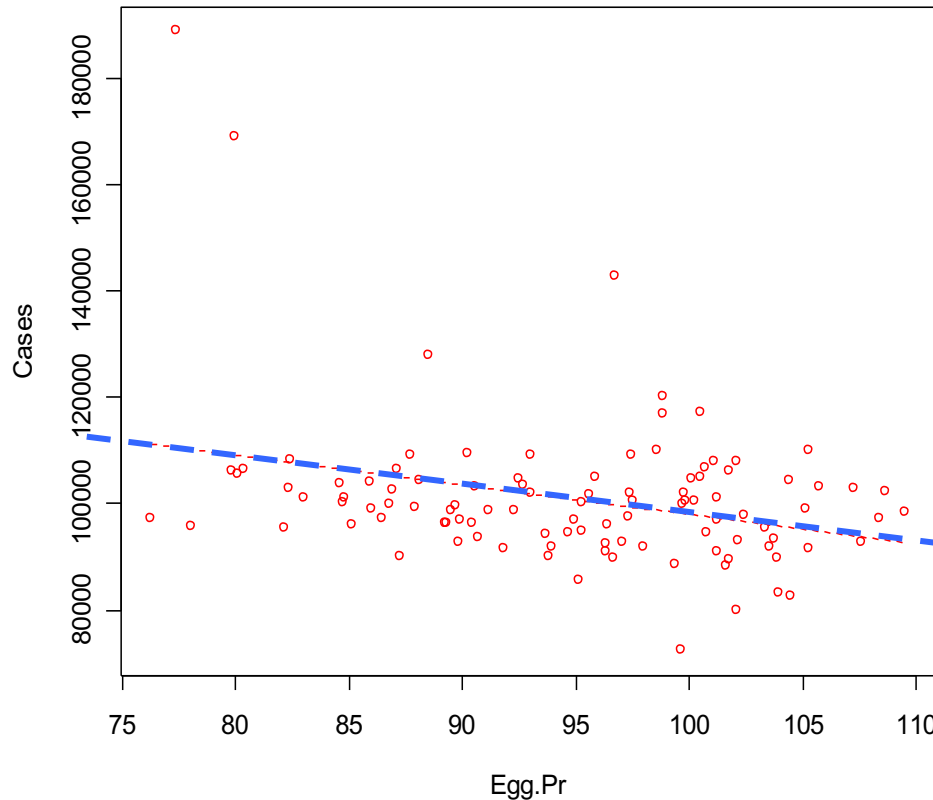
The coefficient of
Egg Price

The estimated standard
deviation of the slope:
Roughly, there is a 95%
chance that the true
slope lies within 2
standard deviations,
i.e., between -217 and -
889

The probability
that the true
slope is zero, i.e.
p-value



Curve Fitting Of The Output



don't use any values of X that aren't contained in the sample data; otherwise the results are subject to a great deal of uncertainty

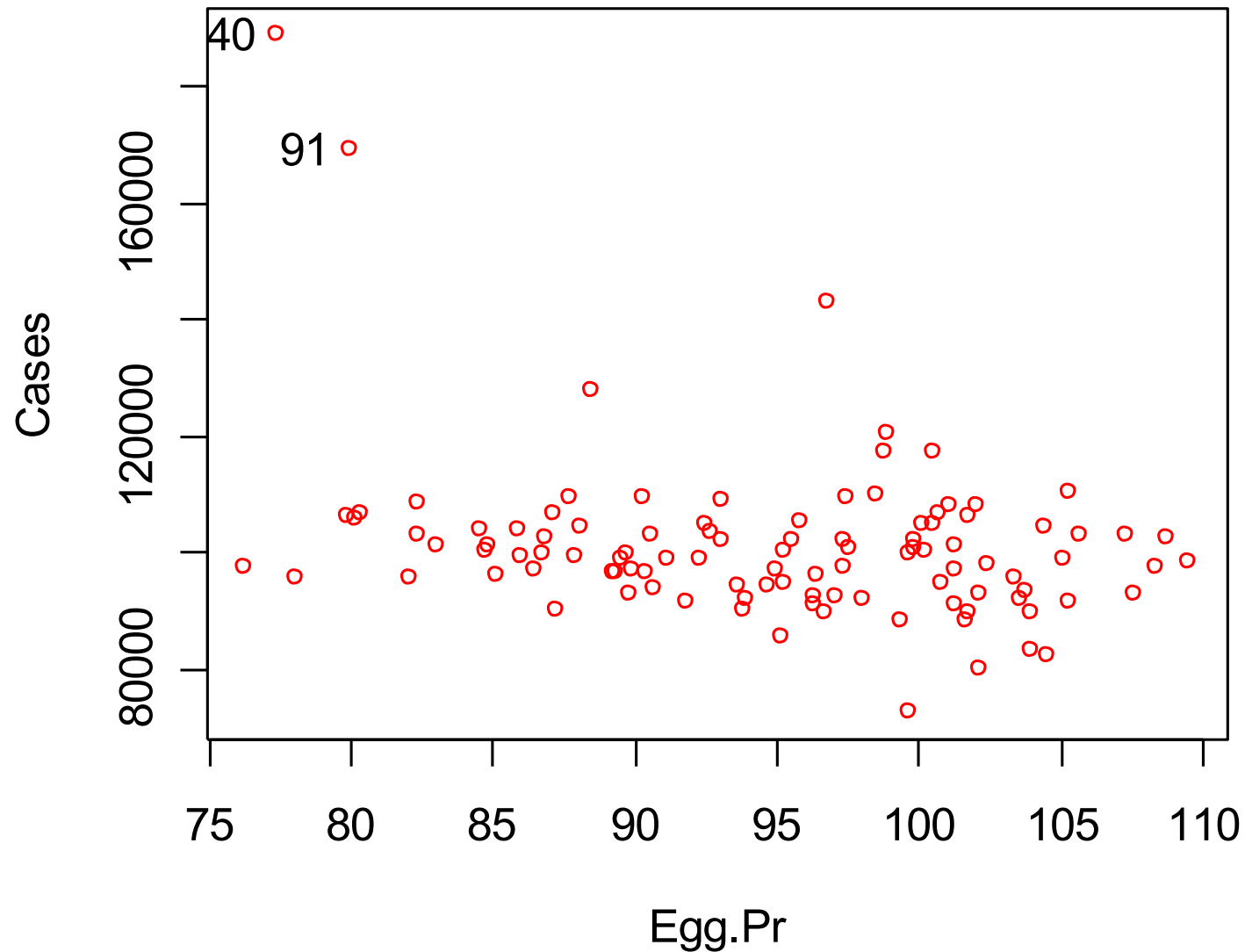
Regression model:

$$\text{Case Sales} = 153414 - 554 \times \text{Egg Price}$$

- ❖ What is the impact of price on the case sales?
- ❖ How are these numbers found?

Minimize **sum of squared errors** between the predicted cases and actual cases is shown as the dashed line

Take Another Look At The Data



Outliers

- ❖ A few sales data points are way outside of the range of the majority of data points
- ❖ We could
 - Drop them
 - Explore to see if there is anything else that predicts them
- ❖ Check: egg sales increase at Easter, and drop immediately after Easter – these weeks (Week 40 & 91) are the outliers

Multiple Linear Regression

- ❖ Simple linear regression: Using ONE independent variable to predict ONE dependent variable
- ❖ Multiple linear regression: Using MORE THAN ONE independent variables to predict ONE dependent variable, e.g. Egg sales can depend not only egg price but also festival (let say Easter)
 - How much increase in sales if it is Easter, or decrease if just after Easter? We need one more independent variable:

$$\text{Egg sales} = \beta_0 + \beta_1 \times \text{Egg Price} + \beta_2 \times \text{Easter} + \varepsilon$$

- ❖ Price is a continuous variable that can be meaningfully multiplied and added. How about Easter? (Categorical variable)

Create Dummy Variables

- ❖ We can use dummy variables to represent categorical variables
- ❖ Variable with n cases can be represented with $n-1$ dummy variables
- ❖ The value without a dummy variable is **base** level, which is indicated with all three dummy variables are zero
 - In this case, the **base level** was **Non Easter**

	Easter	Pre-Easter	Post-Easter
Non-Easter	0	0	0
Easter	1	0	0
Pre-Easter	0	1	0
Post-Easter	0	0	1

- ❖ Our new model:

$$\begin{aligned} & \text{Egg sales} \\ &= \beta_0 + \beta_1 \times \text{Egg Price} + \beta_2 \times \text{Easter} + \beta_3 \times \text{PreEaster} \\ &+ \beta_4 \times \text{PostEaster} + \varepsilon \end{aligned}$$

Interpreting The Output

Coefficients:	Estimate	Pr(> t)
▪ Intercept	115387.19	< 2e-16 ***
▪ Egg Price	-170.15	0.0813
▪ Easter	32728.55	1.94e-08 ***
▪ PreEaster	76946.67	< 2e-16 ***
▪ PostEaster	-22096.43	8.25e-05 ***

➔ Sales prediction function:

$$\text{Sales} = 115387 - 170 \times \text{Egg Price} + 32729 \times \text{Easter} \\ + 76947 \times \text{PreEaster} - 22096 \times \text{PostEaster}$$

❖ To interpret: the coefficient is the comparison to the base case

- The egg sales of Easter week is 32729 more than non-Easter week
- You may try the other two

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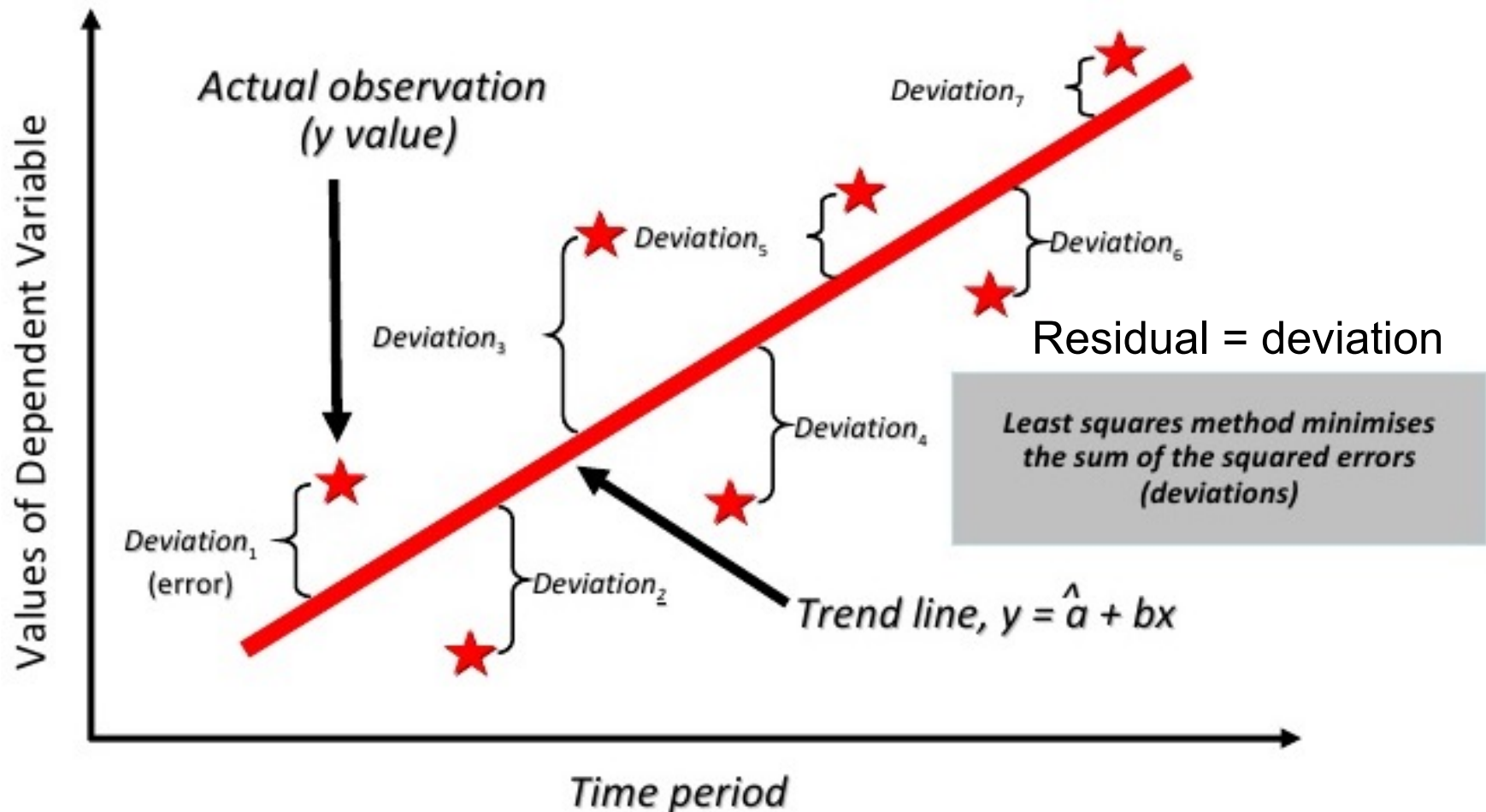
Lecture 2

- ☒ **Predictive Analytics**
 - ☐ Linear Regression
 - ☒ **Parameter Training**
 - ☐ Overfitting

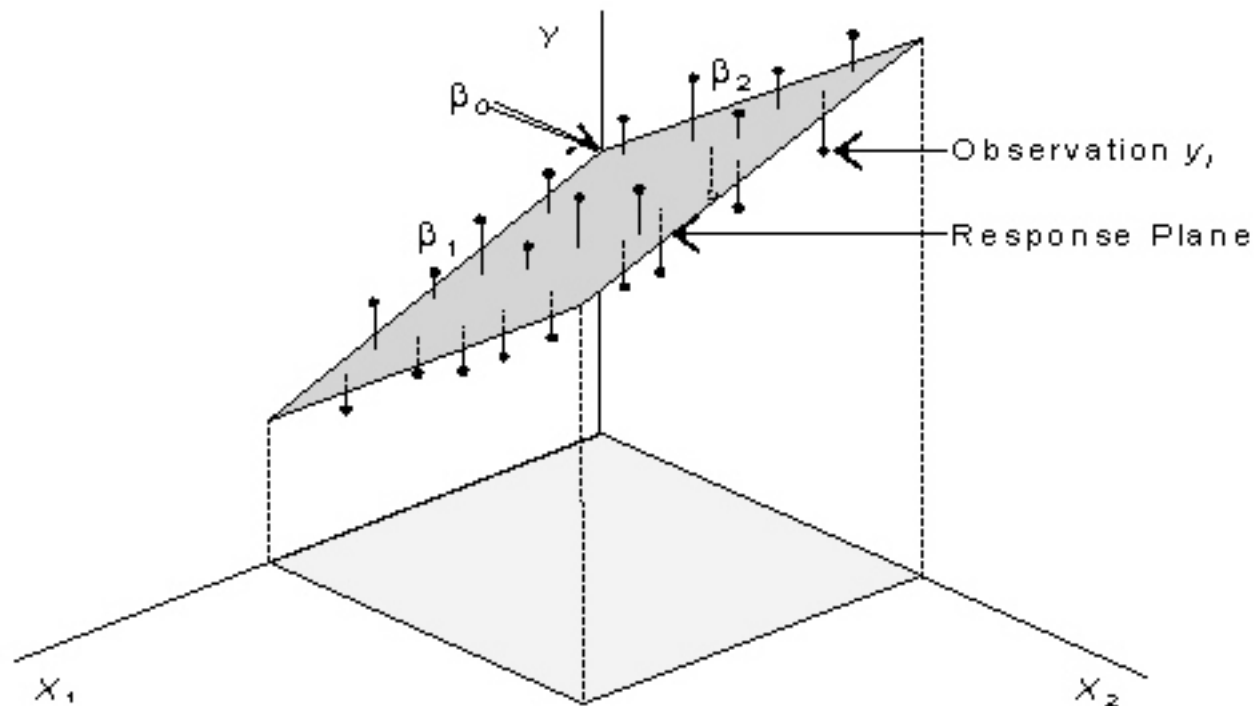
Reading Material:

- ☐ **HM Chapter 4**

Least Sum of Squares Curve Fitting



Multiple Linear Regression



The Linear Model

Scalar Response

Vector of Parameters

Vector of Covariates

Real Value + Noise

$$y = \theta^T x + \epsilon$$

Linear Combination of Covariates

$$\sum_{i=1}^p \theta_i x_i$$

Noise Model:
 $\epsilon \sim N(0, \sigma^2)$

Conditional Likelihood $p(y|x)$

❖ Conditioned on x :

$$y = \overbrace{\theta^T x}^{\text{Constant}} + \epsilon \sim N(0, \sigma^2)$$

Normal Distribution
Mean Variance

❖ Conditional distribution of Y :

$$Y \sim N(\theta^T x, \sigma^2)$$

$$p(y|x) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{(y - \theta^T x)^2}{2\sigma^2} \right)$$

Optimal Parameter

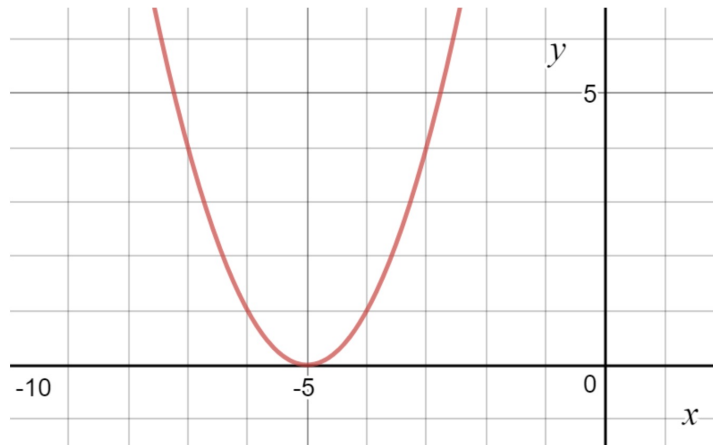
$$y = \theta^T x + \epsilon$$

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta \in \mathbb{R}^p} \sum_{i=1}^n (y_i - \theta^T x_i)^2$$

Minimize Sum (Error)²

Only one variable: Find minimum of $y=(x+5)^2$

$$\rightarrow \frac{dy}{dx} = 2(x + 5) = 0 \rightarrow x = -5$$



Minimizing the Squared Error

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta \in \mathbb{R}^p} \sum_{i=1}^n (y_i - \theta^T x_i)^2$$

❖ Taking the gradient ($\log \mathcal{L}(\theta) = \sum_{i=1}^n (y_i - \theta^T x_i)^2$)

$$-\nabla_{\theta} \log \mathcal{L}(\theta) = \nabla_{\theta} \sum_{i=1}^n (y_i - \theta^T x_i)^2$$

Chain Rule $\rightarrow$ $= -2 \sum_{i=1}^n (y_i - \theta^T x_i) x_i$

$$= -2 \sum_{i=1}^n y_i x_i + 2 \sum_{i=1}^n (\theta^T x_i) x_i$$

Minimizing the Squared Error

- ❖ Rewriting the gradient in matrix form:

$$\begin{aligned} -\nabla_{\theta} \log \mathcal{L}(\theta) &= -2 \sum_{i=1}^n y_i x_i + 2 \sum_{i=1}^n (\theta^T x_i) x_i \\ &= -2X^T Y + 2X^T X \theta \end{aligned}$$

- ❖ Setting gradient equal to 0 and solve for θ_{MLE} :

$$-\nabla_{\theta} \log \mathcal{L}(\theta) = -2X^T y + 2X^T X \theta = 0$$

$$(X^T X) \hat{\theta}_{\text{MLE}} = X^T Y$$

$$\hat{\theta}_{\text{MLE}} = (X^T X)^{-1} X^T Y$$

Normal
Equations

Gradient Descent:

What if p is large? (e.g., $n/2$)

- ❖ Computational inefficiency in calculating

$$\hat{\theta}_{\text{MLE}} = (X^T X)^{-1} X^T Y$$

- ❖ Solution: Iterative Methods

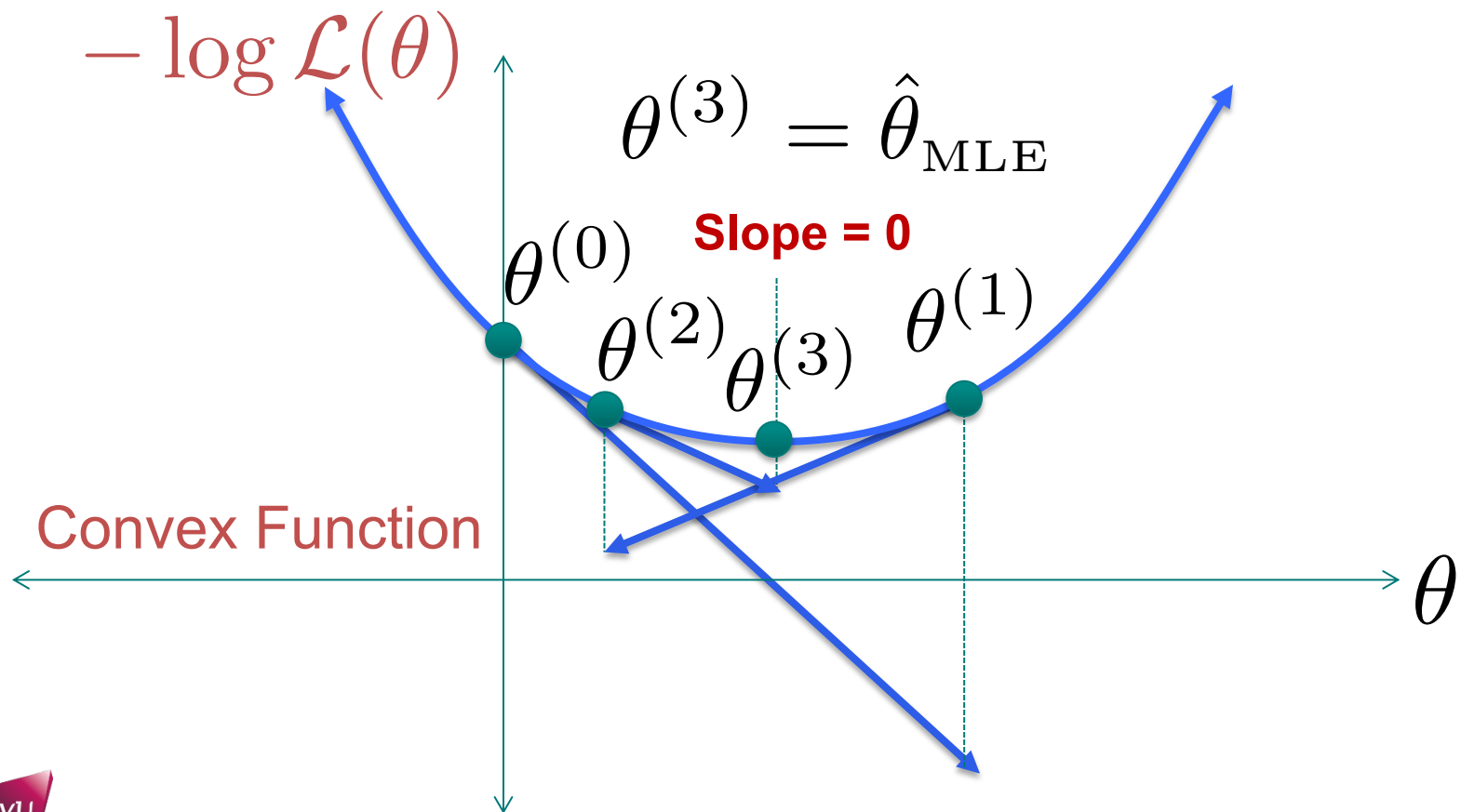
↻ Gradient Descent:

For τ from 0 until *convergence*

$$\theta^{(\tau+1)} = \theta^{(\tau)} - \rho(\tau) \nabla \log \mathcal{L}(\theta^{(\tau)} | D)$$

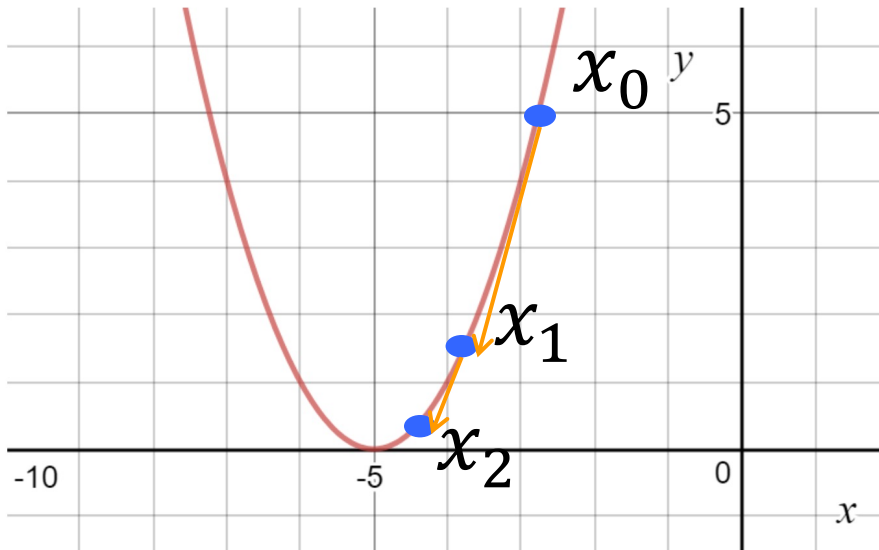
Learning rate

Gradient Descent Illustrated:



Example

Find minimum of $y=(x+5)^2$



$$\frac{dy}{dx} = \frac{d}{dx} (x + 5)^2 = 2 * (x + 5)$$

Initial $x_0 = -3$

Learning rate = 0.2

1. First iteration:

$$x_1 = x_0 - \text{learning rate} * \text{descent}$$

$$x_1 = -3 - 0.2 * (2 * (-3 + 5))$$

$$x_1 = -3.8$$

Question:

1. How many iterations do we need?
2. What is the optimal value?

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Lecture 2

☐ Predictive Analytics

☐ Linear Regression

☐ Parameter Training

☐ Overfitting

Reading Material:

☐ HM Chapter 4

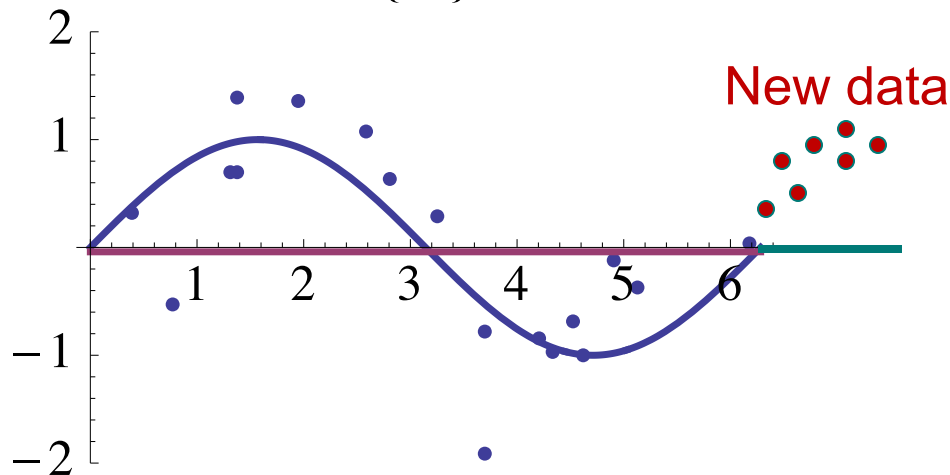
Evolutionary Model Building

- ❖ Predicting the future is based on the assumption that the future repeats the past
- ❖ However, how to evaluate performance for **“future prediction”**?
- ❖ How many IVs provide the best performance for “future” prediction?

Underfitting VS Overfitting

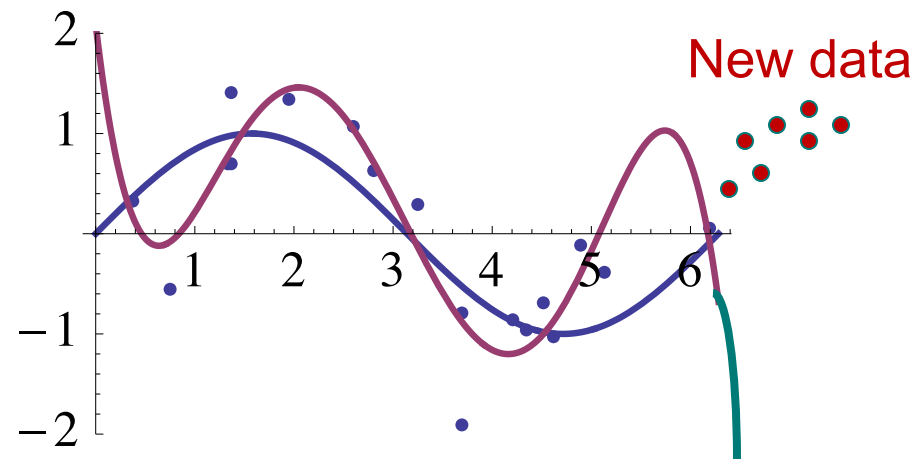
Under-fitting

{1.}



Large training error

Over-fitting



Small training error
Large testing error

Using Validation Set

❖ Divide training data into two parts:

❧ Training set:

❖ use for model building

❧ Validation set:

❖ use for estimating generalization error

❖ Note: validation set is not the same as test set

❖ Drawback:

❧ Less data available for training

Model Evaluation

Hold-out method

3) Use D_v to select the model class which has smallest empirical error on D_v

$$\hat{\lambda} = \arg \min_{\lambda \in \Lambda} \hat{R}_V(\hat{f}_\lambda)$$

└─ Evaluated on validation dataset D_v

4) Hold-out predictor

$$\hat{f} = \hat{f}_{\hat{\lambda}}$$

Intuition: Small error on one set of data will not imply small error on a randomly sub-sampled second set of data

Model Evaluation

Hold-out method

We would like to pick the model that has smallest generalization error.

Can judge generalization error by using an independent sample of data.

Hold - out procedure:

n data points available $D \equiv \{X_i, Y_i\}_{i=1}^n$

1) Split into two sets: Training dataset Validation dataset **NOT test Data !!**
 $D_T = \{X_i, Y_i\}_{i=1}^m$ $D_V = \{X_i, Y_i\}_{i=m+1}^n$

2) Use D_T for training a predictor from each model class:

$$\hat{f}_\lambda = \arg \min_{f \in \mathcal{F}_\lambda} \hat{R}_T(f)$$

└─→ Evaluated on training dataset D_T

Example – Hold out

Dataset

X	Y
1.0	3.3
2.9	9.8
3.3	10.3
4.9	14.8
5.6	16.8
6.3	19.8
7.2	21.8
8.3	25.2
9.7	29.6
10.5	31.8
11.5	35.1
12.8	38.7
13.0	39.9
14.7	44.2
15.7	47.8
16.2	49.0
17.5	52.9
18.6	56.7
19.4	58.5
20.2	61.2

X	Y
1.0	3.3
2.9	9.8
3.3	10.3
4.9	14.8
5.6	16.8
6.3	19.8
7.2	21.8
8.3	25.2
9.7	29.6
10.5	31.8
11.5	35.1
12.8	38.7
13.0	39.9
14.7	44.2
15.7	47.8
16.2	49.0
17.5	52.9
18.6	56.7
19.4	58.5
20.2	61.2

Randomly select
80% as training



Training error

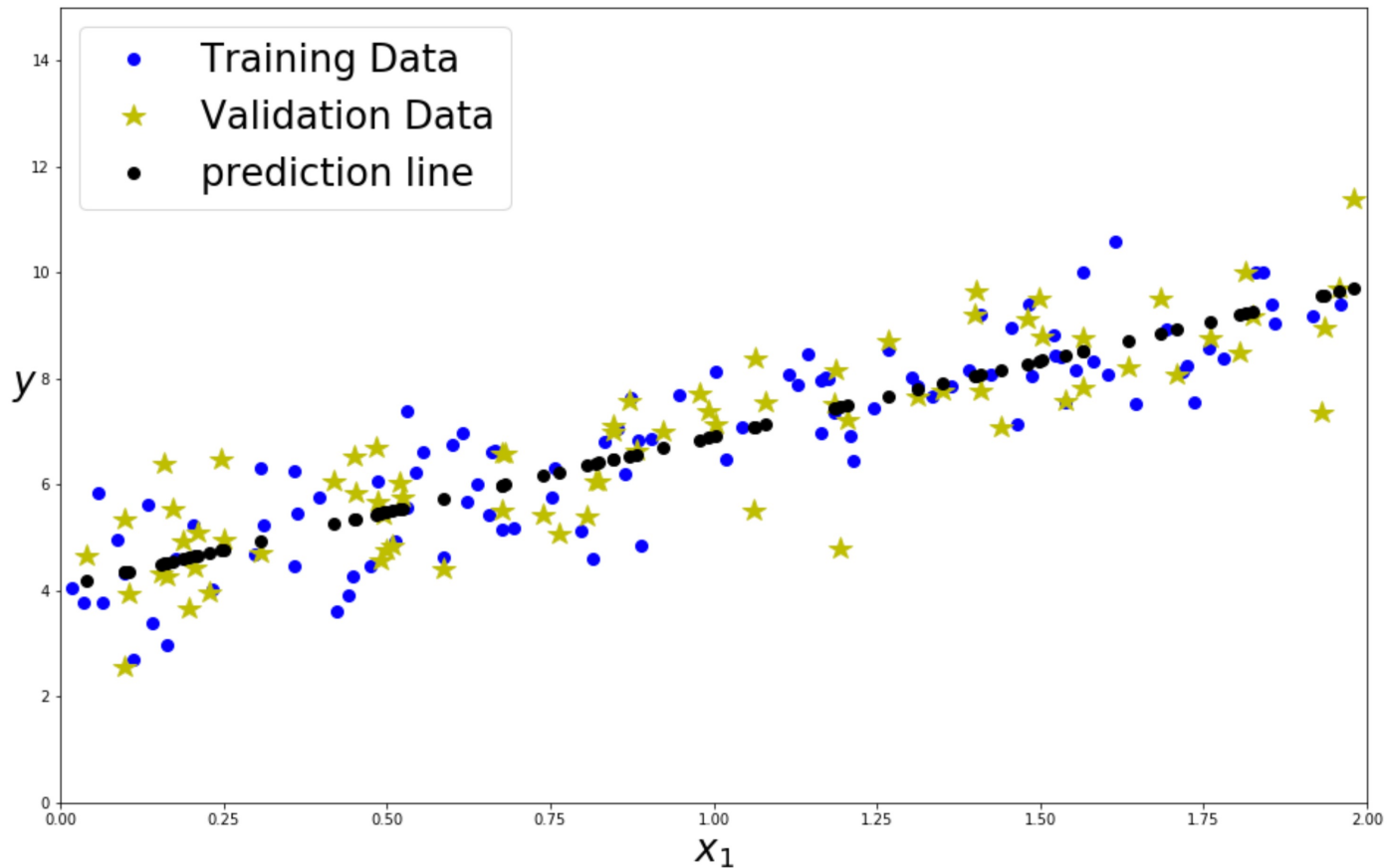
Rest 20% as
validation



Validation error

Example

$$y = 3 * x + 4 + \epsilon$$



Regularization

- ❖ Straightforward method to prevent overfitting!
- ❖ How it works?
 - ☞ We make some constraints when training our model!

Regularization

❖ Recall the training process:

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta \in \mathbb{R}^p} \sum_{i=1}^n (y_i - \theta^T x_i)^2$$

Objective: minimize the MSE:

$$\min \sum_{i=1}^n (y_i - (a_1 x_1 + a_2 x_2 + \cdots a_j x_j + b))^2$$

❖ Regularization: training with constraints.

Objective: minimize the MSE:

$$\min \sum_{i=1}^n (y_i - (a_1 x_1 + a_2 x_2 + \cdots a_j x_j + b))^2$$

subject to: $R(a) \leq r$

Regularization

❖ Regularization: training with constraints.

Objective: minimize the MSE:

$$\min \sum_{i=1}^n (y_i - (a_1 x_1 + a_2 x_2 + \cdots a_j x_j + b))^2$$

subject to: $R(a) \leq r$

❖ Equivalent to

$$\min \sum_{i=1}^n (y_i - \theta^T x_i)^2 + \lambda * R(a)$$

λ : Regularization parameter $\lambda \geq 0$

Regularization - Lasso

$$\diamond R(a) = \sum_{j=1}^p |a_j|$$

$$\min \sum_{i=1}^n (y_i - (ax + b))^2 + \lambda * \sum_{j=1}^p |a_j|$$

- ❖ If $\lambda = 0$, Lasso regression is Linear Regression
- ❖ The regularization may force some parameter $a_j = 0$ if it has a big variance, so as to reduce model complexity.
- ❖ If λ is too large - $\rightarrow$ force all $a_j = 0$

Regularization - Ridge

$$\diamond R(a) = \sum_{j=1}^p |a_j|^2$$

$$\min \sum_{i=1}^n (y_i - (a_1x_1 + a_2x_2 + \cdots a_jx_j + b))^2 + \frac{\lambda}{2} * \sum_{j=1}^p |a_j|^2$$

Target

Prediction result

Training Error

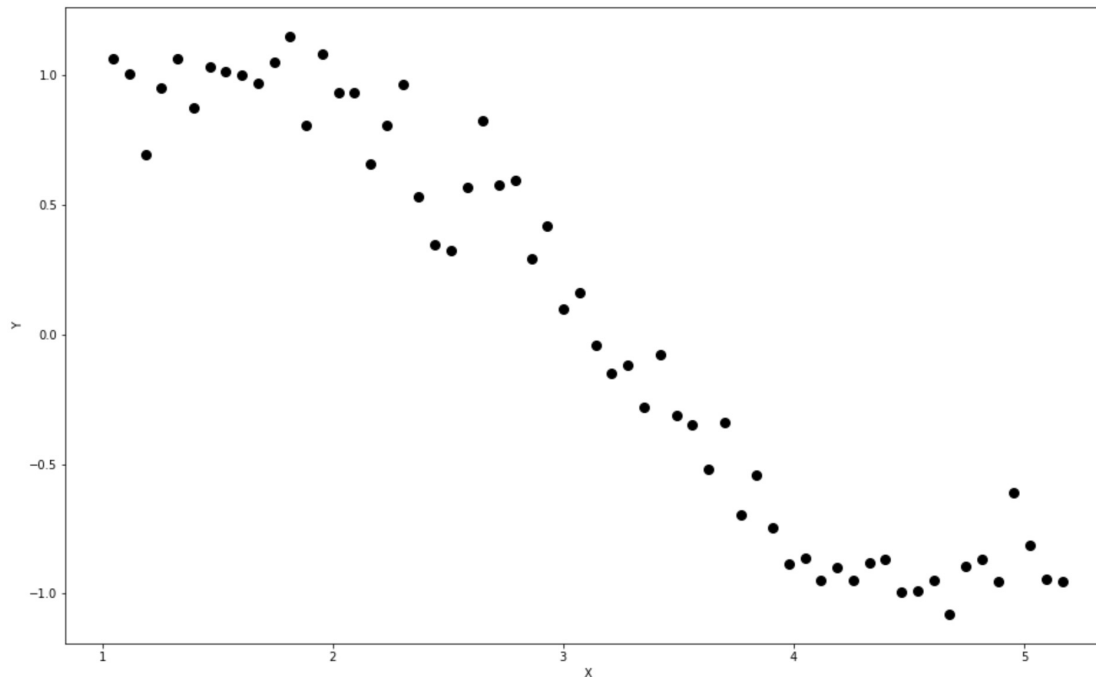
Regularization
Penalty on model
complexity



Force some
coefficients to be 0

Example

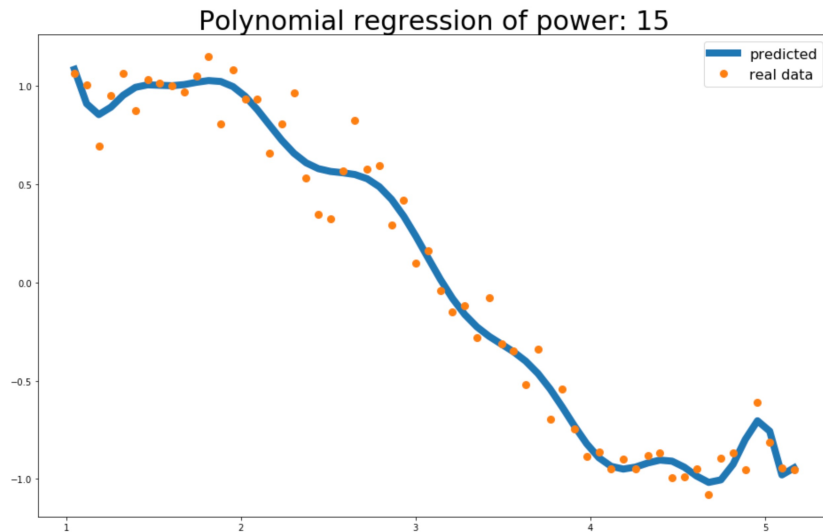
The data to be fitted by polynomial regression:



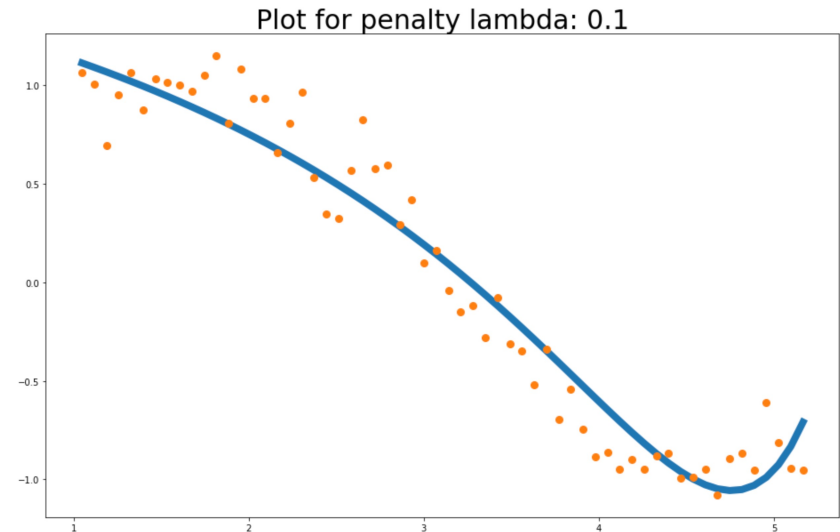
$$y = a_0 + a_1 * x + a_2 * x^2 + a_3 * x^3 + \dots a_{15} * x^{15}$$

Fitting result

Simple linear regression



Ridge regression



Model	intercept	a1	a2	a3	a4	a5	a6	a7
Simple Linear Regression	-36247.81543	243862.2	-745882.84	1376243.5	-1713742.3	1526233.6	-1004821.3	498314.29
Ridge Regression	1.392827	-0.224699	-0.033668	-0.005179	-0.000733	-0.000091	-0.000009	0

Model	a8	a9	a10	a11	a12	a13	a14	a15
Simple Linear Regression	-187821.78	53847.777	-11656.844	1872.8394	-216.36711	16.983958	-0.810681	0.017753
Ridge Regression	0	0	0	0	0	0	0	0

Take Away For Today

- ❖ Predictive Modeling
- ❖ Simple and multiple linear regression
- ❖ Interpretation of the output: coefficient, p-value
- ❖ Dummy variables for categorical variables
- ❖ Interpretation of dummy variable coefficient
- ❖ Evaluate model fitness
- ❖ Independent variable selection